

How to do the ADF Test for each column of my Dataframe in Python? [closed]

Asked 4 years, 11 months ago Modified 17 days ago Viewed 3k times

▲

2

▼

🔖

🕒

Closed. This question does not meet [Stack Overflow guidelines](#). It is not currently accepting answers.

💡

This question does not appear to be about programming within the scope defined in the [help center](#).

Closed 17 days ago.

Improve this question

I am looking for a way to do the ADF Test for multiple columns of a DataFrame. Do you have any idea?

This is my multi-column dataframe. How to calculate the test for each column separately?

Date	Acre	Amapá	Pará	Rondônia	Roraima	Tocantins	Alagoas	Bahia	Ce
2010-01-01	152.77	162.36	149.31	150.82	175.02	141.55	138.12	141.45	14
2010-02-01	154.45	165.34	150.93	152.84	177.48	143.83	133.67	142.29	14
2010-03-01	155.94	164.94	151.49	155.76	179.39	145.53	126.44	143.66	14
2010-04-01	158.96	167.52	152.70	158.56	181.25	147.02	123.97	145.03	14
2010-05-01	161.59	167.76	154.24	161.21	180.97	149.28	124.92	146.89	14
2019-08-01	185.74	216.15	185.76	165.51	256.45	194.93	138.88	163.24	18
2019-09-01	186.86	216.95	187.16	166.02	258.57	195.44	145.83	163.69	18
2019-10-01	185.86	218.30	187.86	166.14	259.87	196.04	146.89	163.63	18
2019-11-01	184.44	219.05	188.65	165.89	262.84	195.93	147.44	164.00	18
2019-12-01	183.22	217.67	187.17	165.18	262.29	194.29	146.98	162.95	18

python

pandas

dataframe

time-series

data-science

Share

Improve this question

Follow

edited Mar 16 at 8:42

karel

5,985625960

asked Apr 3, 2021 at 23:23

Alex Gois

5715

4 Answers

Sorted by:

Highest score (default)

⬆



1



Not sure about your data in particular but you can use the statsmodels library for that. Here is the docs about the ADF test : <https://www.statsmodels.org/stable/generated/statsmodels.tsa.stattools.adfuller.html>

Share Improve this answer Follow

answered Apr 4, 2021 at 0:08



benja d

156 1 8



! Sign up to request clarification or add additional context in comments.



Comments

Add a comment



This is a function you can apply to columns of DataFrame:

1



```
from statsmodels.tsa.stattools import adfuller
import pandas as pd
```



```
def adfuller_test(series, signif=0.05):
    """
```



```
    Perform Augmented Dickey-Fuller to test for Stationarity of the given series
    and print report. Null Hypothesis: Data has unit root and is non-stationary.
```

```
    series: time series in pd.Series format
    signif: significance level for P-value to reject Null Hypothesis
    """
```

```
        x = adfuller(series, autolag='AIC')
```

```
        #using dictionary saves different data types (float, int, boolean)
```

```
        output = {'Test Statistic': x[0],
                  'P-value': x[1],
                  'Number of lags': x[2],
                  'Number of observations': x[3],
                  f'Reject (signif. level {signif})': x[1] < signif }
```

```
        for key, val in x[4].items():
            output[f'Critical value {key}'] = val
```

```
    return pd.Series(output)
```

You can follow like `df_train.apply(lambda x: adfuller_test(x), axis=0)` to get a summary.

Share Improve this answer

edited Aug 10, 2021 at 14:09

answered Aug 10, 2021 at 14:04

Follow



Valerii

11 2

Comments

Add a comment



```
result = adfuller(prices, autolag='AIC')
```

```
print(f'ADF Statistic: {result[0]}')
print(f'p-value: {result[1]}')
print(f'n_lags: {result[2]}')
print(f'n_obs: {result[3]}')
```

```
for key, value in result[4].items():
    print('Critical Values:')
    print(f'    {key}: {value}')
```

OR

```
def adf_test(series, name):
    result = adfuller(series.dropna())
    print(f'--- {name} ---')
    print(f'  ADF Statistic: {result[0]:.4f}')
    print(f'  p-value:         {result[1]:.4f}')
    print(f'  Stationary:      {"YES" if result[1] < 0.05 else "NO - unit root"}')
    print(f'  Critical values: {result[4]}')
    print()

adf_test(prices, 'Price levels')
adf_test(daily_returns, 'Daily returns')
```

Share Improve this answer

edited Mar 16 at 4:55

answered Mar 16 at 4:51

Follow



Mohar Chaudhuri

1 1

1 Comment

Add a comment



Community Mar 16 at 5:18

As it's currently written, your answer is unclear. Please [edit](#) to add additional details that will help others understand how this addresses the question asked. You can find more information on how to write good answers [in the help center](#).

2

Reply





0



```
import statsmodels.tsa.stattools as tsa
```

```
df = ... #load your dataframe  
adf_results = {}
```

```
for col in df.columns.values: #or edit this for a subset of columns first  
    adf_results[col] = tsa.adfuller(df[col])
```

You can use the library statsmodels from python with the function <https://www.statsmodels.org/dev/generated/statsmodels.tsa.stattools.adfuller.html>

Share Improve this answer Follow

answered Mar 16 at 6:53



PV8

6,374

9

54

114

Comments

Add a comment

Start asking to get answers

Find the answer to your question by asking.

Ask question

Explore related questions

python

pandas

dataframe

time-series

data-science

See similar questions with these tags.